MIL-P-45570B(AL)
27 August 1985
SUPERSEDING
MIL-P-45570A(WC)
28 March 1974

MILITARY SPECIFICATION

PLUG SET, RADIATOR TEST

This specification is approved for use by the Armament Research and Development Center (Tool and Equipment Engineering Directorate, Standardization and Specification Engineering Division), Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1 SCOPE

1.1 This specification covers one adjustable compression-type motor vehicle radiator test plug set for plugging the radiator hose inlet(s) and outlet(s) during tests and repairs.

2 APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM) 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Industry association specifications and standards are generally available for reference from Libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3 REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more complete radiator test plug set(s) for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Materials. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the automotive maintenance industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3 Design. The plug set shall be new and of the manufacturer's latest design meeting the requirements as specified herein. All parts subject to wear or replacement shall be accessible.

3.4 Construction. All parts and materials used in fabrication of the plug set shall be new and free of defects. In no event shall patching processes be used for reclaiming defective parts. All surfaces of all metal parts shall be clean and free of extraneous materials.

* 3.5 Performance and product characteristics. The sealing element portion of each plug in the plug set shall be capable of recovering from stress periods of compression, as experienced during radiator testing, without visual evidence of compression set or aging. All metal parts of the plug set shall be made of brass, stainless steel, bronze or copper. Each radiator plug, when installed for testing, shall be capable of withstanding a 20 pound-force per square inch gage (psig) pressure for a period of 1 minute without leakage.

3.6 Sealing element range. The test plug set shall provide the capability of plugging holes from 7/8 inch diameter to 2-1/2 inch diameter when sealing the inlet and outlet of the radiator. The set shall include 1-1/4 inch, 1-1/2 inch, 1-5/8 inch, 1-3/4 inch, 2 inch, 2-1/8 inch, 2-1/4 inch, 2-3/8 inch and 2-5/8 inch sizes respectfully. Two plugs of each size shall be furnished to allow concurrent plugging of the inlet and outlet.

* 3.7 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.8 Identification marking. Each plug set shall be marked for identification in accordance with the MIL-STD-130 and unless otherwise specified (see 6.2.1) shall include the National Stock Number.

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3.9 Workmanship. Standards of workmanship shall be such as to assure equipment of a neat general appearance which complies with the requirements of section 3.

4 QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection of production items (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3 and the test under 4.4. Failure of the first article to pass the examination or the test shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance on the contract or purchase order. As a minimum, quality conformance inspection and test shall consist of "a" and "b" as follows. Sampling will be conducted in accordance with MIL-STD-105 single sampling for normal inspection at the inspection level and Acceptable Quality Level (AQL) specified.

a. Product examination (see 4.3). Sampling for examination shall be conducted in accordance with General Inspection Level II. Acceptance will be based on an AQL of 1.0 for major defects and 2.5 for minor defects.

b. Inspection of packaging (see 4.5). Sampling for inspection shall be conducted in accordance with Special Inspection Level S-3. Acceptance will be based on AQL of 4.0.

4.2.2.1 Lot. All items of the same type presented for acceptance at the same time shall be considered a lot for the purpose of inspection and test.

4.3 Product examination. Production items shall be examined to determine compliance with the requirements of the contract. As a minimum, the product characteristics contained in 4.3.1 shall be examined.

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* 4.3.1 Product characteristics.

List of defects	Classification of defects Major	Minor
Design not as specified (see 3.3)	X	
Construction not as specified (see 3.4)	X	
Sealing element range not as specified (see 3.6)	X	
Threads not as specified (see 3.7)	X	
Identification marking not as specified (see 3.8)		X
Workmanship not as specified (see 3.9)		X

4.4 Pressure test. The assembled test plugs shall be inserted into either a radiator hose-connection, or a mock-up test chamber with openings to allow the test plugs to be installed for the test. The test plugs shall withstand a 20 psig pressure for a period of 1 minute. While under pressure, check for leakage around the test plug. Vent the pressure and remove the test plugs. Examine the test plugs for conformance with the requirements of this specification (see 3.5).

* 4.5 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5 PACKAGING

* 5.1 Preservation. Preservation shall be level A, or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each component of the plug set shall be cleaned by process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.2 Unit package. Plug sets shall be individually preserved-packaged method III of MIL-P-116. The unit container shall conform to grade W6c of PPP-B-636. Instructions issued with the plug set shall be preserved-packaged method IC-1 of MIL-P-116 and placed inside the unit container. The unit container shall be closed by taping in accordance with method IV in the appendix to the container specification.

* 5.1.2 Industrial. The plug sets shall be preserved-packaged in accordance with the requirements of ASTM D3951-82.

* 5.2 Packing. Packing shall be level A, level B, or Industrial, as specified (6.2.1).

5.2.1 Level A. The plug sets preserved-packaged as specified (see 5.1), shall be packed in a container conforming to PPP-B-621, class 2 or PPP-B-601 (overseas type). The quantity packed shall be governed by the size and weight limitations of the applicable container specification. Closure and strapping shall be in accordance with the applicable container specification and the appendix thereto.

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5.2.2 Level B. The plug sets preserved-packaged as specified (see 5.1), shall be packed in a container conforming to PPP-B-621, class 1; PPP-B-601 (domestic type); or PPP-B-640, class 2. The quantity packed shall be governed by the size and weight limitations of the applicable container specification. Closure and strapping shall be in accordance with the applicable container specification and the appendix thereto.

* 5.2.3 Industrial. The plug sets shall be packed in accordance with the requirements of ASTM D3951-82.

5.3 Marking.

5.3.1 Military markings. When military levels of protection are specified, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and any special markings required by the contract or order.

* 5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of ASTM D3951-82.

6 NOTES

6.1 Intended use. The radiator test plug set covered by this specification is used to plug motor vehicle-type radiator inlet and outlet hose connections prior to submerging the radiator into a water tank to discover the existence of leaks in the radiator core or radiator tank being repaired.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, when required (see 3.1).

c. Identification marking, if different (see 3.8).

d. First article inspection, if different (see 4.2.1).

e. Selection of applicable level of preservation-packaging (see 5.1).

f. Selection of applicable level of packing (see 5.2).

6.2.2 Contract data requirements. Required technical data such as operating instructions, parts lists, and other instructions for operation and maintenance, as identified on a number DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

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6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-45570B(AL)	Plug Set, Radiator Test

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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